

Lecture 5

Programmable Logic Devices (PLD's)

Introduction

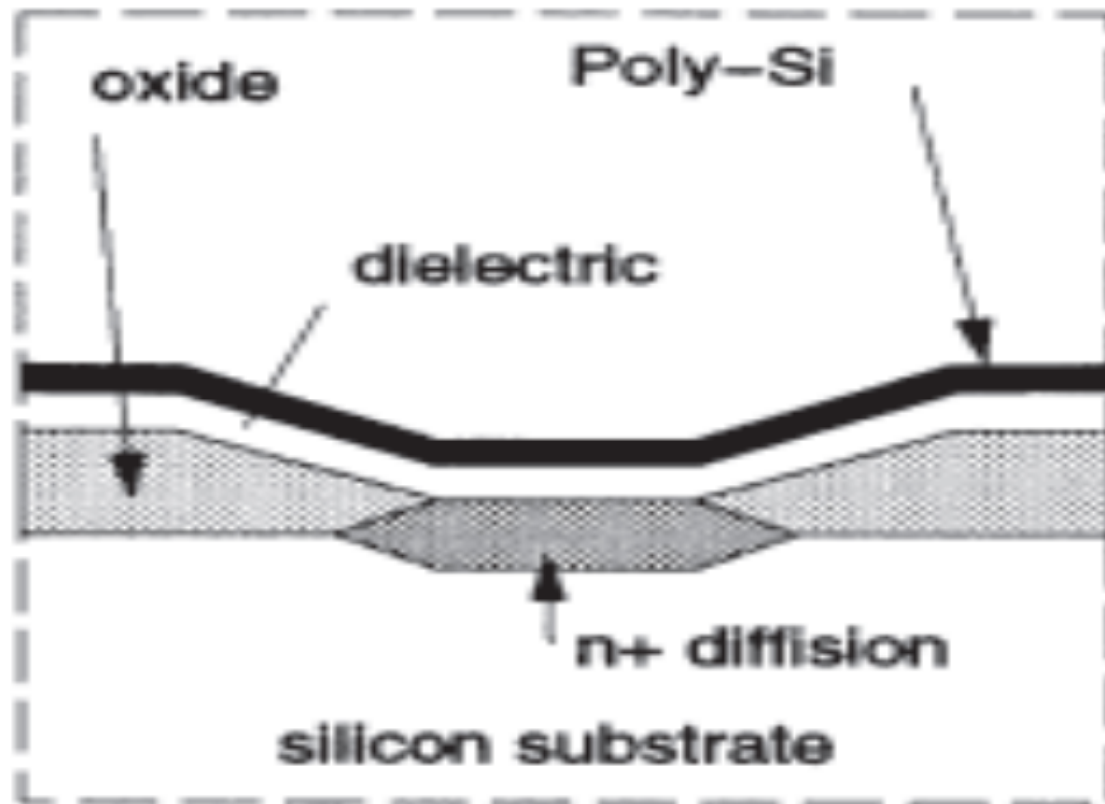
- A typical **programmable logic device** may have hundreds to millions of gates interconnected through hundreds to thousands of internal paths.
- Programmable logic devices (PLDs) are fabricated with structures that implement logic functions and structures that are used to control connections or to store information specifying the actual logic functions implemented.
- These latter structures require programming, a hardware procedure that determines which functions are implemented.

Programming Technologies:

1- Permanent Connection

- **Fuses** which are initially CLOSED are selectively “blown out” by a higher than normal voltage to established OPEN connections.
- **Anti-fuses**, the opposite of fuses, contain a material that is initially non-conducting (OPEN). Anti-fuses are selectively CLOSED by applying a higher-than-normal voltage to provide a pattern of OPEN and CLOSED anti-fuses to define the logic.
- **Mask programming**, which is done by the semiconductor manufacturer during the last steps of the chip fabrication process.

Anti-Fuses



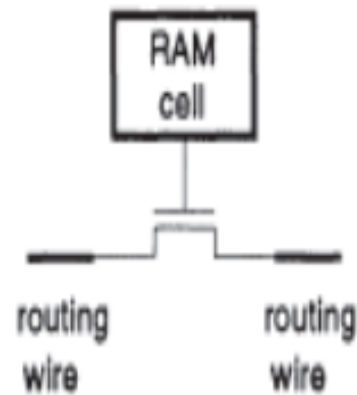
2- Reprogrammed Connections

- **SRAM Transistor**. Since storage element content can be changed electronically, the device can be easily reprogrammed.
- But in order to store values, the power supply must be available.
- **EPROM technology**. This popular technology is based on storing charge on a floating gate.
- Stored negative charge on the floating Gate makes the transistor impossible to turn ON.
- The absence of stored negative charge makes it possible for the transistor to turn ON if a HIGH is applied to its regular gate.
- Since it is possible to add or remove the stored charge, these technologies can permit erasure and reprogramming.

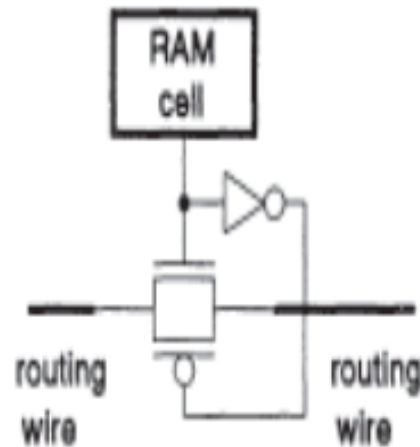
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- Two approaches using control of transistor switching are called **erasable** and **electrically erasable**.
- Programming applies combinations of voltage higher-than normal power-supply voltages to the transistor.
- Erasure uses exposure to a strong ultraviolet light source for a specified amount of time.
- Once this type of chip has been erased, it can be reprogrammed.
- An electrically erasable device can be erased by a process somewhat similar to the programming process.
- Flash technology is a form of electrically erasable technology that has a variety of erase options, including the erase of stored charge from individual floating gates, all of the floating gates, or specific subsets of floating gates.

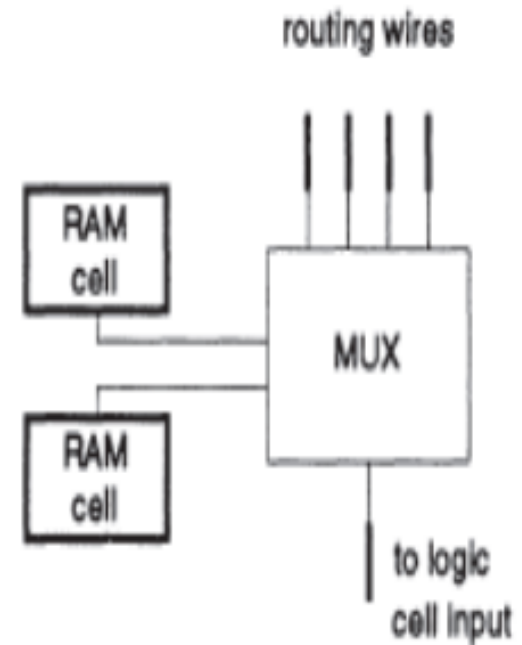
SRAM Programming Technologies



a) pass-transistor

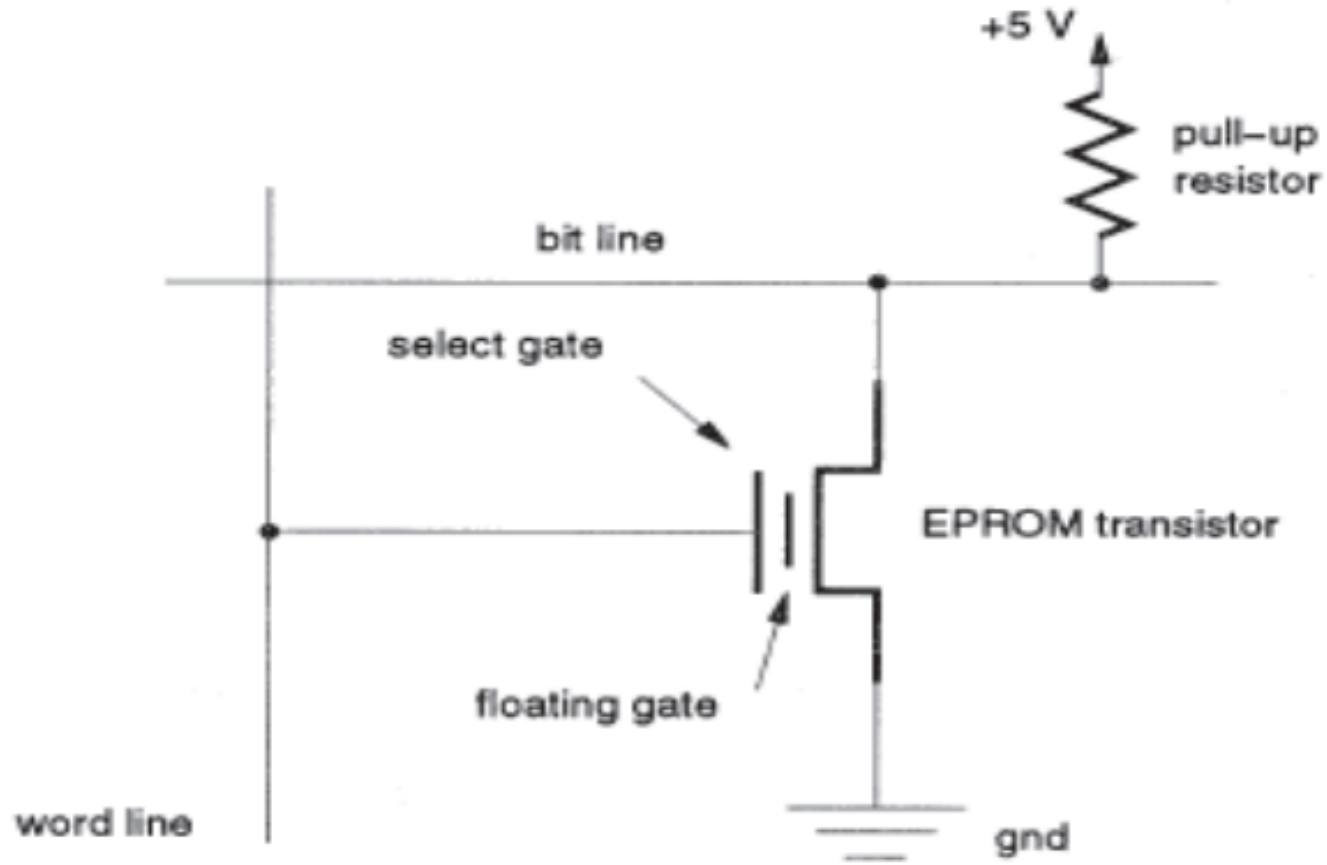


b) transmission gate



c) multiplexer

EPROM Programming Tech.,



Combinational PLDs

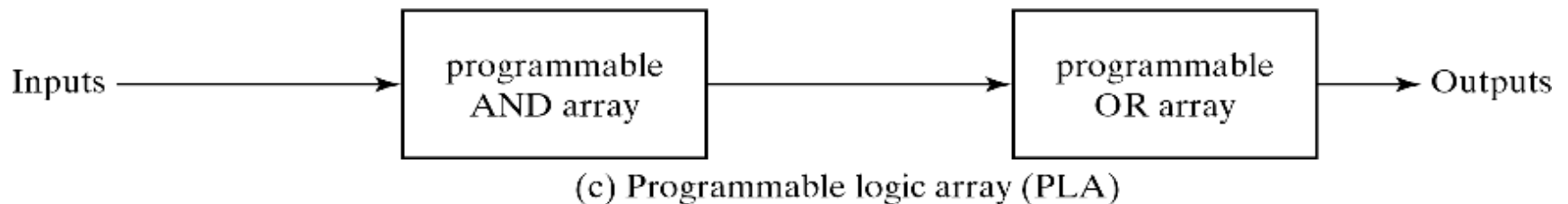
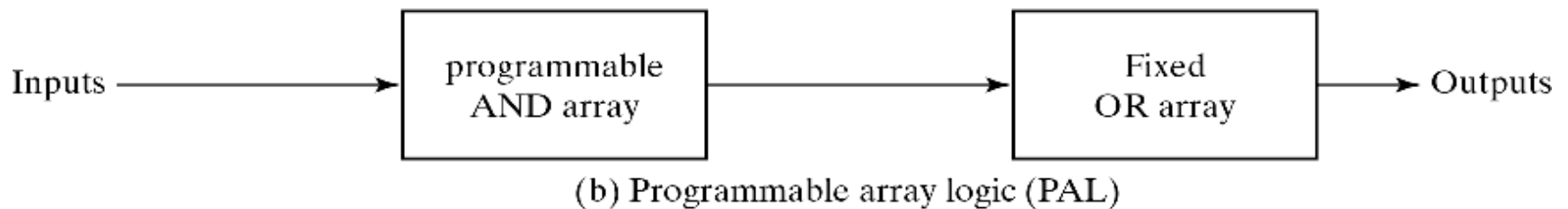
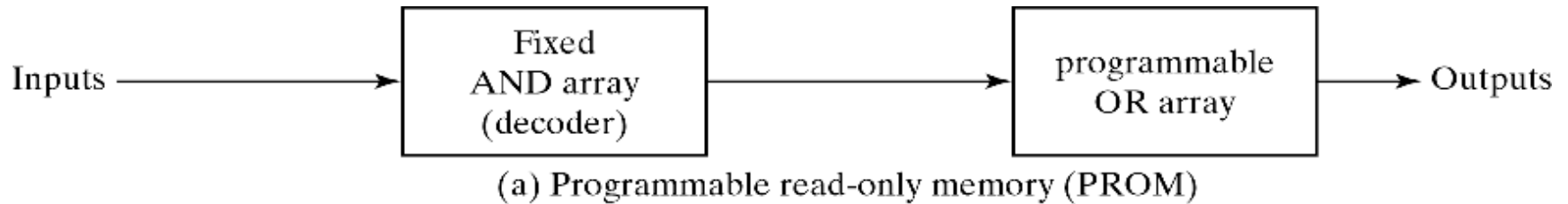


Fig. 7-13 Basic Configuration of Three PLDs

1- Read-Only Memory

- A block diagram of a ROM is shown below.
- It consists of k address inputs and n data outputs.
- The number of words in a ROM is determined from the fact that k address input lines are needed to specify 2^k words.

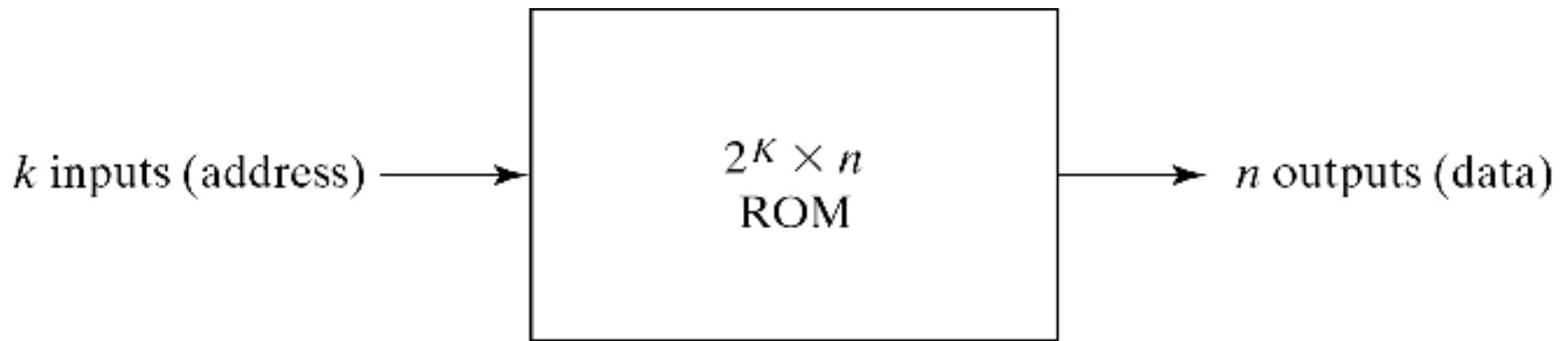


Fig. 7-9 ROM Block Diagram

Construction of ROM

- Each output of the decoder represents a memory address.
- Each OR gate must be considered as having 32 inputs.
- A $2^k \times n$ ROM will have an internal $k \times 2^k$ decoder and n OR gates.

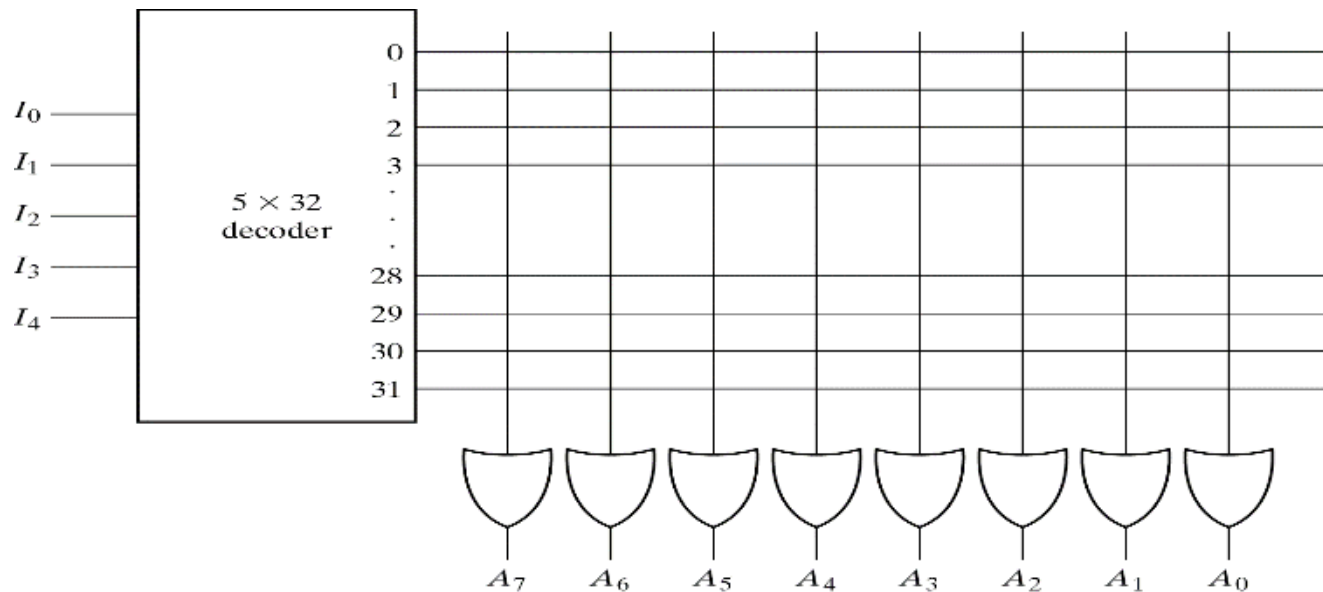


Fig. 7-10 Internal Logic of a 32×8 ROM

Truth table of ROM

- A programmable connection between two lines is logically equivalent to a switch that can be altered to either be close or open.
- Intersection between two lines is sometimes called a cross-point.

Table 7-3
ROM Truth Table (Partial)

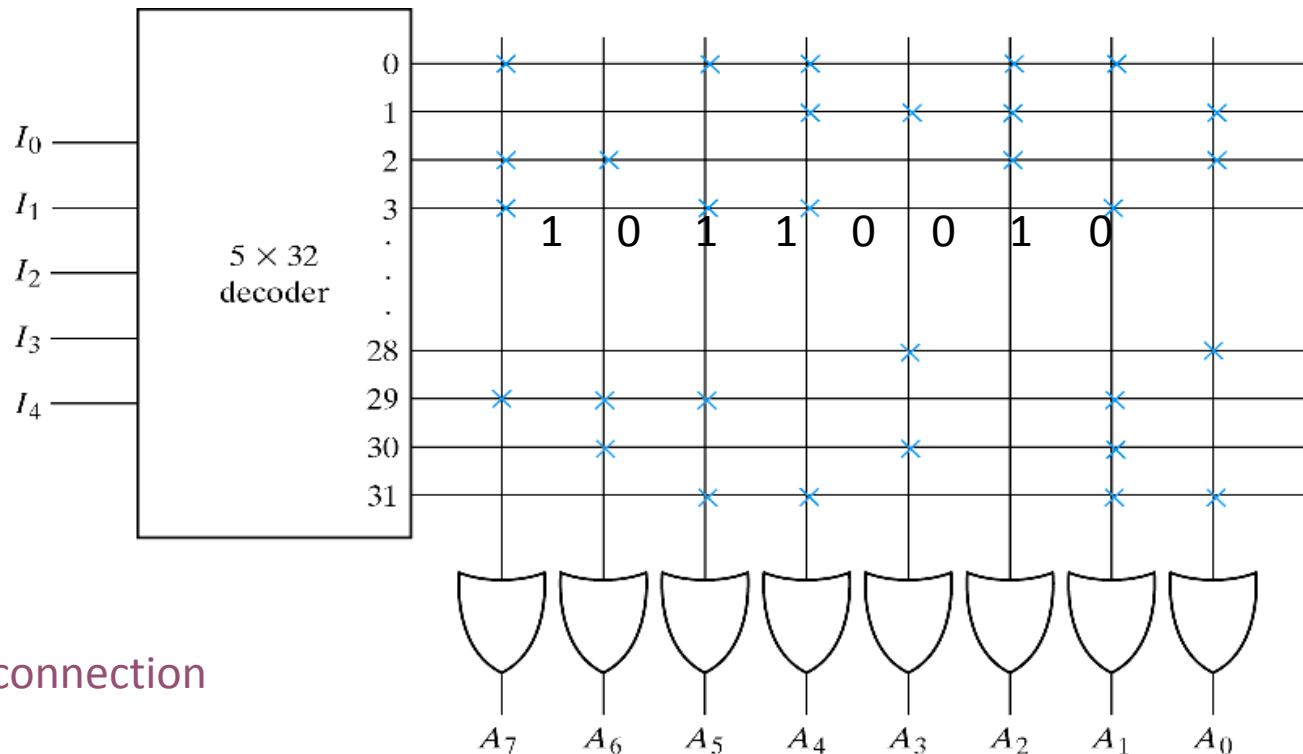
Inputs					Outputs							
I4	I3	I2	I1	I0	A7	A6	A5	A4	A3	A2	A1	A0
0	0	0	0	0	1	0	1	1	0	1	1	0
0	0	0	0	1	0	0	0	1	1	1	0	1
0	0	0	1	0	1	1	0	0	0	1	0	1
0	0	0	1	1	1	0	1	1	0	0	1	0
		⋮					⋮					
1	1	1	0	0	0	0	0	0	1	0	0	1
1	1	1	0	1	1	1	1	0	0	0	1	0
1	1	1	1	0	0	1	0	0	1	0	1	0
1	1	1	1	1	0	0	1	1	0	0	1	1

Programming the ROM

In Table 7-3, 0 ☐ no connection

1 ☐ connection

Address 3 = 10110010 is permanent storage using fuse link



X : means connection

Fig. 7-11 Programming the ROM According to Table 7-3

Combinational circuit implementation

- The **internal operation** of a ROM can be interpreted in two way:
- **First**, a memory unit that contains a fixed pattern of stored words.
- **Second**, implements a combinational circuit.
- Fig. 7-11 may be considered as a combinational circuit with eight outputs, each being a function of the five input variables.

$$A_7(I_4, I_3, I_2, I_1, I_0) = \Sigma(0,2,3...,29)$$

Types of ROMs

- Depending on the programming technology and approaches, read-only memories have different names:
 1. ROM—mask programmed,
 2. PROM—fuse or anti-fuse programmed,
 3. EPROM—erasable floating gate programmed,
 4. EEPROM or E2 PROM—electrically erasable floating gate programmed,
 5. and FLASH Memory—electrically erasable floating gate with multiple erasure and programming modes.

2- Programmable Logic Array (PLA)

- The decoder in PROM is replaced by an array of AND gates that can be programmed to generate any product term of the input variables.
- The product terms are then connected to OR gates to provide the sum of products for the required Boolean functions.
- The output is inverted when the XOR input is connected to 1 (since $x \oplus 1 = x'$). The output doesn't change and connect to 0 (since $x \oplus 0 = x$).

Programming Table

1. First: lists the product terms numerically
2. Second: specifies the required paths between inputs and AND gates
3. Third: specifies the paths between the AND&OR gates
4. For each output variable, we may have a T(ure) or C(complement) for programming the XOR gate

Example

$$F1 = AB' + AC + A'BC'$$

$$F2 = (AC + BC)'$$

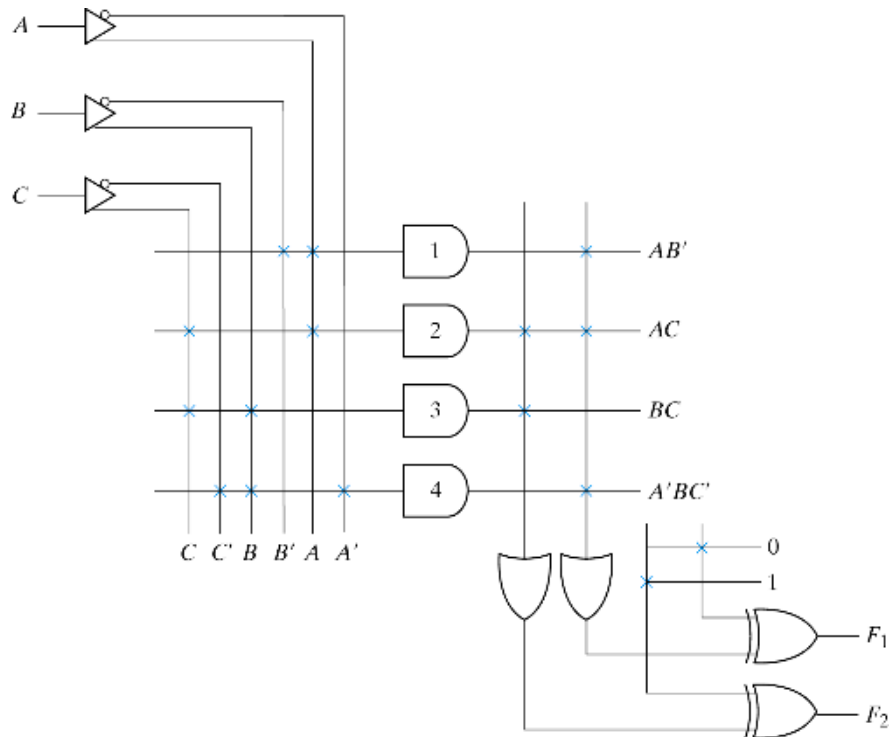


Fig. 7-14 PLA with 3 Inputs, 4 Product Terms, and 2 Outputs

Table 7-5
PLA Programming Table

Product Term		Inputs			Outputs	
					(T)	(C)
		A	B	C	F_1	F_2
AB'	1	1	0	-	1	-
AC	2	1	-	1	1	1
BC	3	-	1	1	-	1
A'BC'	4	0	1	0	1	-

Disadvantages of combinational PLD's

- With their simple two-level structure, both types of PLDs described above allow high speed-performance implementations of logic circuits.
- However, the simple structure also leads to their main drawback.
- They can only implement small logic circuits that can be represented with a modest number of product terms, because their interconnection structure would grow impractically large if the number of product terms were increased.

Sequential Programmable Devices

1. Sequential (or simple) Programmable Logic Device (**SPLD**)
2. Complex Programmable Logic Device (**CPLD**)
3. Field Programmable Gate Array (**FPGA**)

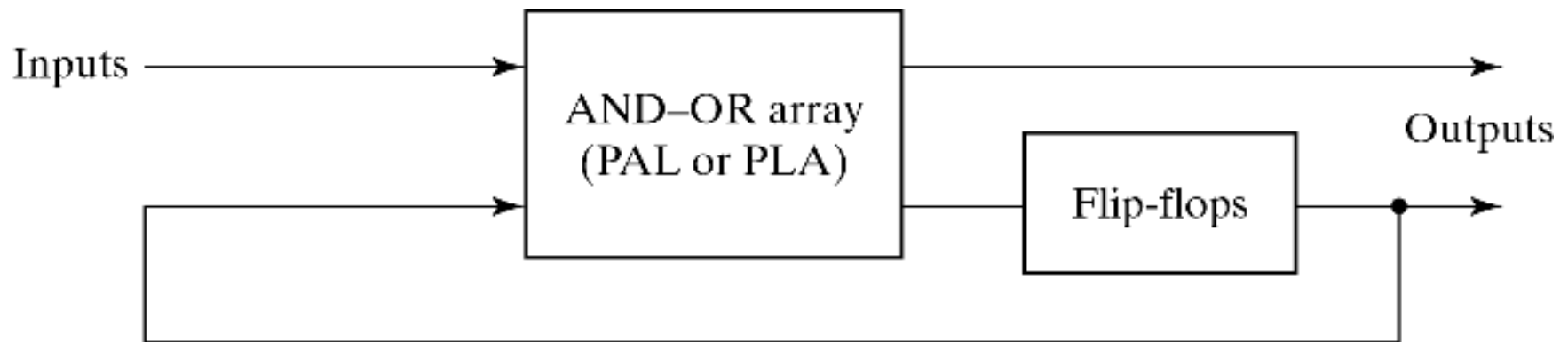


Fig. 7-18 Sequential Programmable Logic Device

CPLD Configuration

- It consists of multiple PLDs interconnected through a programmable switch matrix.

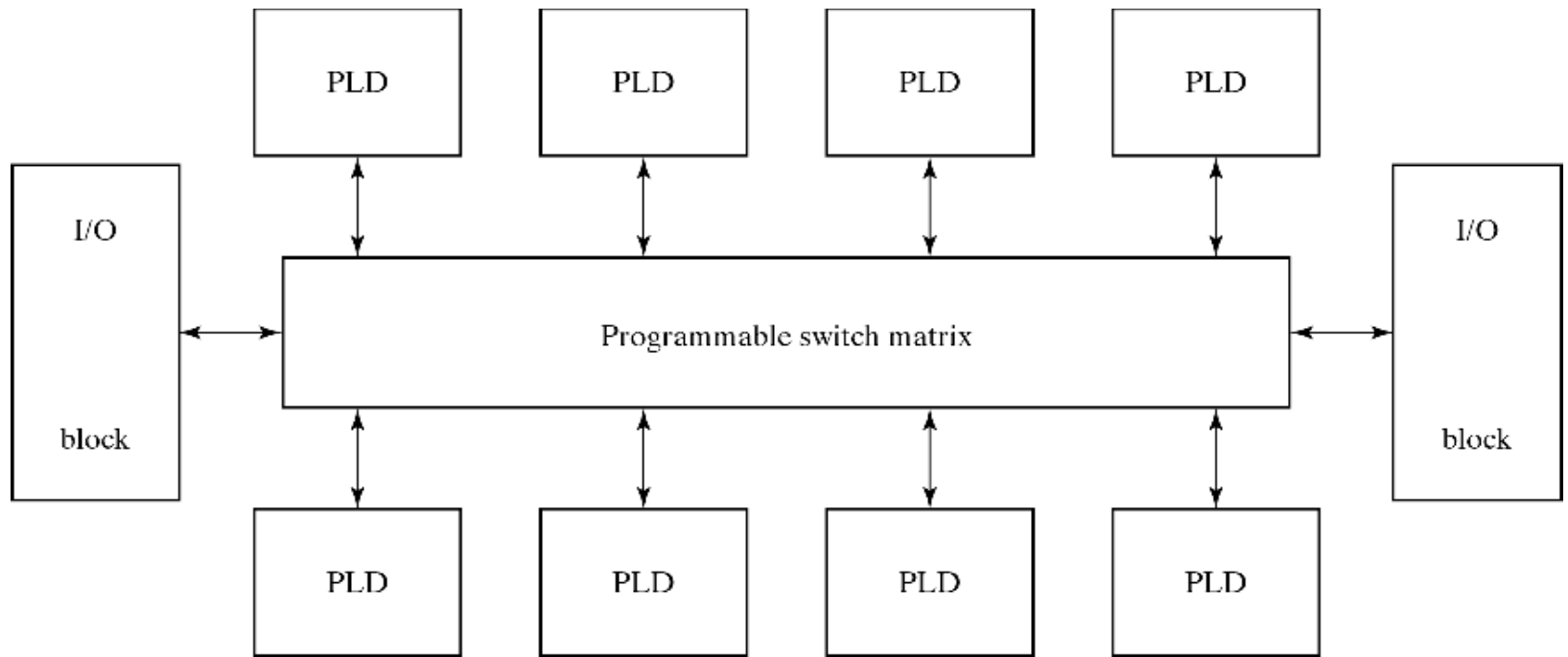


Fig. 7-20 General CPLD Configuration

Gate Array

- The basic component used in VLSI design is the gate array.
- A gate array consists of a pattern of gates fabricated in an area of silicon that is repeated until the entire chip is covered with the gates.
- Arrays of gates are fabricated within a single IC chip depending on the technology used.

MPGA

- The most general type of programmable devices consists of an array of uncommitted elements that can be interconnected according to a user's specifications.
- Such is the class of devices known as *Mask-Programmable Gate Arrays* (MPGAs).
- The most popular MPGAs consist of rows of transistors that can be interconnected to implement a desired logic circuit.

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- User specified connections are available both within the rows (to implement basic logic gates) and between the rows (to connect the basic gates together).
- In addition to the rows of transistors, some circuitry is provided that handles input and output to the external pins of the IC package.

FPGA

- FPGA is a VLSI circuit that can be programmed in the user's location.
- FPGA consists of an array of uncommitted elements that can be interconnected in a general way.
- the interconnections between the elements are user-programmable.
- A typical FPGA logic block consists of look-up tables, multiplexers, gates, and flip-flops.
- Look-up table is a truth table stored in a SRAM and provides the combinational circuit functions for the logic block.

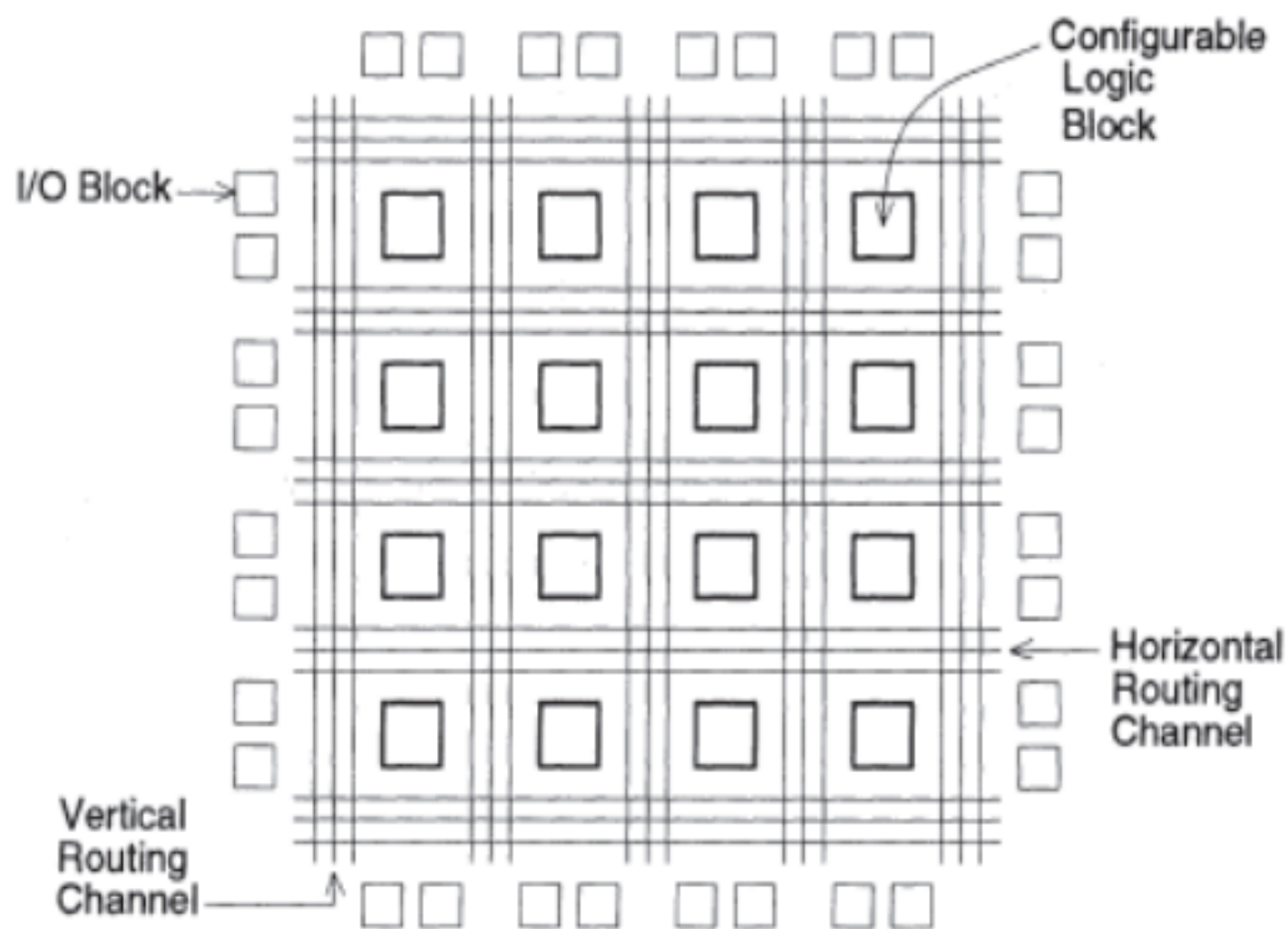
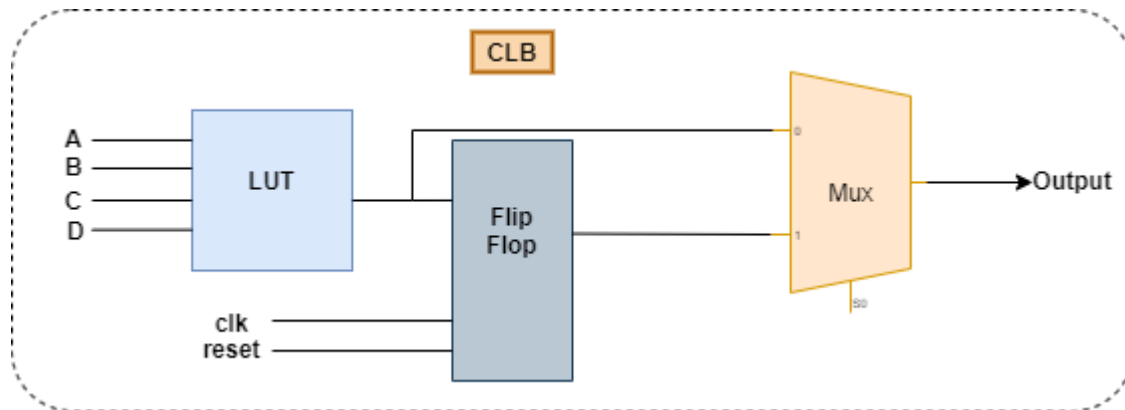
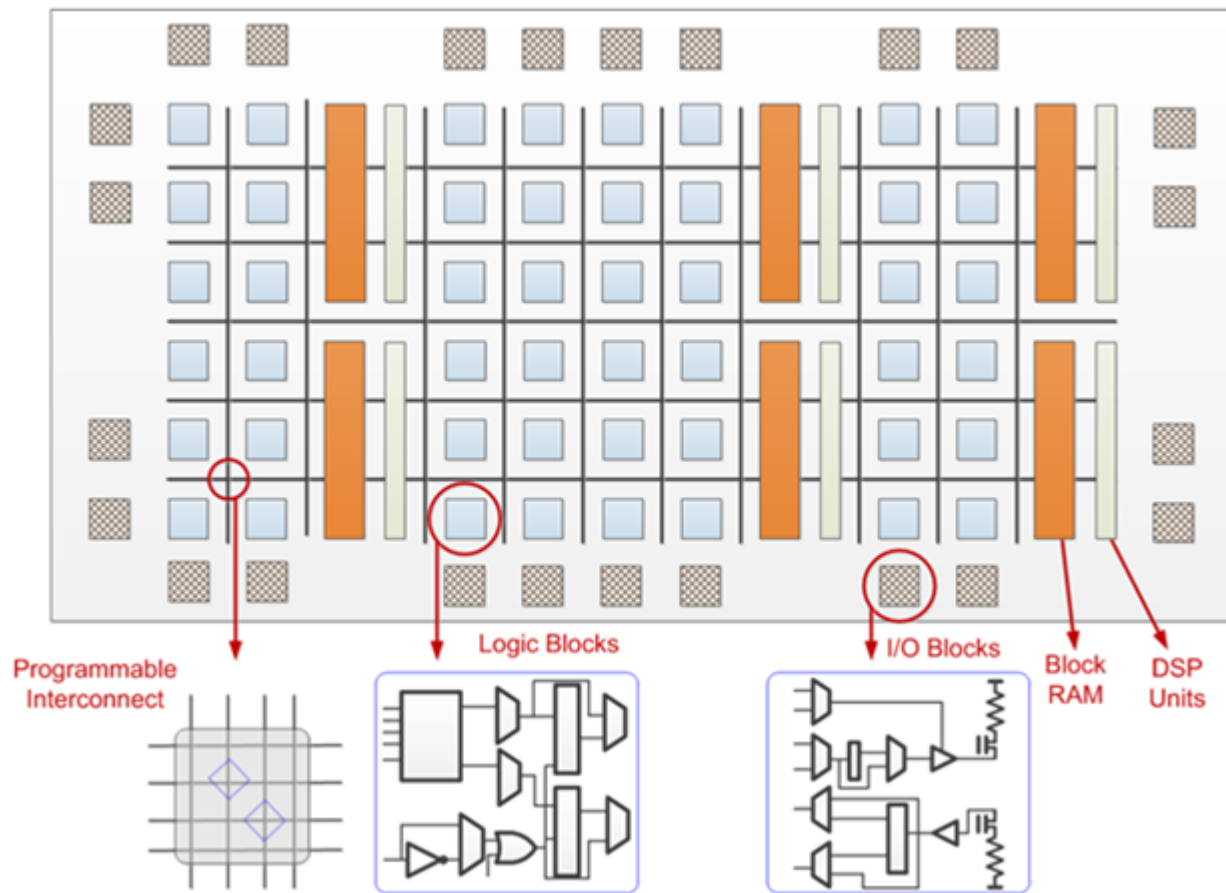
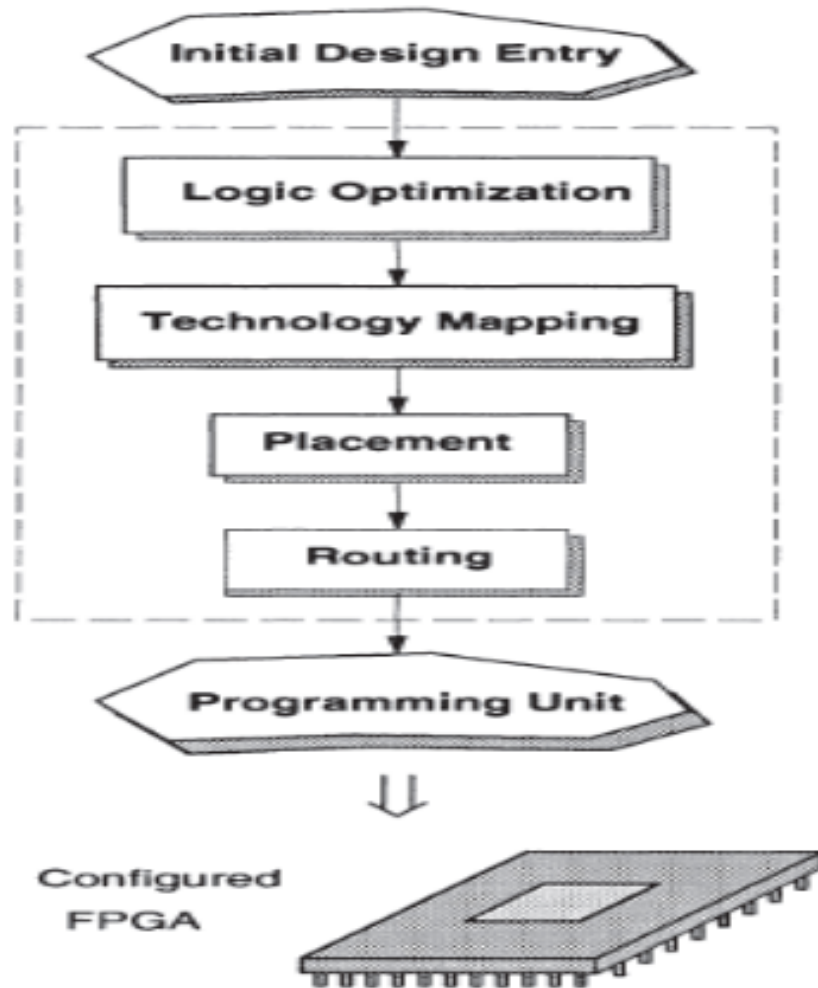


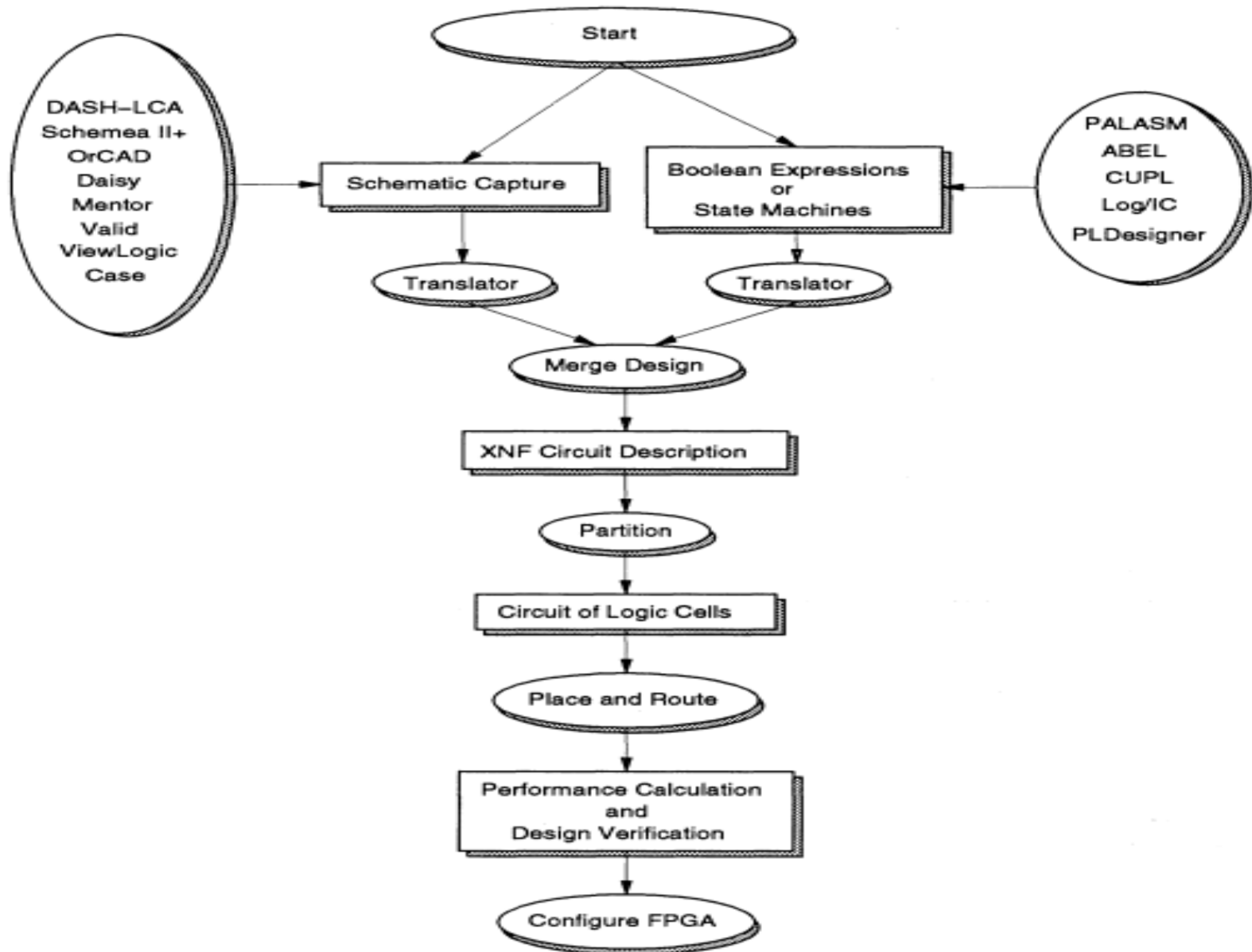
Figure 2.6 - General Architecture of Xilinx FPGAs.



Implementation Process



Xilinx Design Flow



Cont.,

- After the logic circuit has been fully designed and merged into one circuit, it is translated into a special format that is understood by Xilinx CAD tools.
- This format is called the Xilinx Netlist Format, or XNF.
- The translation utility is supported either by Xilinx or by the vendor of the logic entry tool.
- The translation process may also involve automatic optimizations of the circuit.
- The XNF circuit is next partitioned into Xilinx Logic Cells. Note that the word partition is used by Xilinx, but the more common term for this step is technology mapping.

Logic Optimization and Technology Mapping

- The original logic network is first manipulated by a *logic optimization* program, which produces an optimized network that is functionally equivalent to the original network.
- Following logic optimization, the technology mapping phase transforms the optimized network into a circuit that consists of a restricted set of circuit elements.
- In the FPGA environment, the circuit elements are the FPGA's logic blocks.

Logic Optimization (miss II Alg.,)

- In the misII synthesis system the complexity of a network is measured by counting the number of literals in the local function for each node.
- Each local function is a sum-of-products expression and each instance of a variable counts as one literal.
- For example, the following 4-input, 2-output network has 11 literals

$$f=ac+ad+bc+bd,$$

$$g=a+b+c$$

- The complexity of this network can be reduced by the following modifications.
- The expression $(a + b)$ is factored out of the equations for nodes f and g , and a new node e , implementing the function $a + b$, is created.
- The variable e is then substituted back into the equations for nodes f and g , resulting is the following 7 literal network.

$$e=a+b$$

$$f=e(c+d)$$

$$g=e +c$$

Placement & Routing

- The next step in such systems is the *placement* of the logic blocks.
- The routing of interconnections among the logic blocks is realized.
- Routing is the final phase of a circuit's implementation, after which the FPGA can be configured by a programming unit.

Modern FPGA Architecture

- The modern-day FPGA architecture emphasizes the more and more LUTs in a smaller area with minimum power utilization, having more flexible on-chip memory.
- They are also focused on putting the adders/multipliers and DSP logic inside the CLB to reduce the latency, faster computation, reduced routing, and increased throughput.
- For many years 4-input LUT is the traditional LUT architecture design in practice.
- Xilinx has adopted a different approach to design a CLB that contains the two slices; each slice has four 6-input LUTs with two outputs and integrated with fast look-ahead carry logic and provide with better routing switches.

Assignment

- **Write about:**
 - **Logic Optimization algorithm**
 - **Technology mapping algorithm**
 - **Routing for Symmetrical FPGAs**